the heat accumulated from sunlight in the air surrounding the telescope. At nightfall, with the dome open, the air temperature near the telescope rapidly equals the temperature of the nighttime air, allowing light from stars and other cosmic objects to be sharp and clear. And, while not directly named for a cosmic object, titanium derives from the Titans of Greek mythology; Titan is Saturn's largest moon.

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By many measures, iron ranks as the most important element in the universe. Massive stars manufacture elements in their core, in sequence from helium to carbon to oxygen to nitrogen, and so forth, all the way up the Periodic Table to iron. With twenty-six protons and at least as many neutrons in its nucleus, iron's odd distinction comes from having the least total energy per nuclear particle of any element. This means something quite simple: if you split iron atoms via fission, they will absorb energy. And if you combine iron atoms via fusion, they will also absorb energy. Stars, however, are in the business of making energy. As high-mass stars manufacture and accumulate iron in their cores, they are nearing death. Without a fertile source of energy, the star collapses under its own weight and instantly rebounds in a stupendous a supernova explosion, outshining a billion suns for more than a week.

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The soft metal gallium has such a low melting point that, like cocoa butter, it will liquefy on contact with your hand. Apart from this parlor demo, gallium is not interesting to astrophysicists, except as one of the ingredients in the gallium chloride experiments used to detect elusive neutrinos from the Sun. A huge (100-ton) underground vat of liquid gallium chloride is monitored for any collisions between neutrinos and gallium nuclei, turning it into germanium. The encounter emits a spark of X-ray light that is measured every time a nucleus gets slammed. The long-standing solar neutrino problem, where fewer neutrinos were detected than predicted by solar theory, was solved using "telescopes" such as this.

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Every form of the element technetium is radioactive. Not surprisingly, it's found nowhere on Earth except in particle accelerators,